that the light approaching the president of Backwardland would be moving more quickly, having received a “kick” from the forward-moving train. Similarly, these observers would see that the light approaching the president of Forwardland would be moving more slowly, being “dragged” back by the train’s motion. When these (erroneous) effects were considered, the observers on the platform would see that that the light beams reached each president simultaneously. However, in the real world light does not speed up or slow down, it cannot be kicked to a higher speed or dragged to a slower one. Platform observers will therefore justifiably claim that the light reached the president of Forwardland first.

The constancy of the speed of light requires that we give up the age-old notion that simultaneity is a universal concept that everyone, regardless of their state of motion, agrees upon. The universal clock previously envisioned to dispassionately tick off identical seconds here on earth and on Mars and on Jupiter and in the Andromeda galaxy and in each and every nook and cranny of the cosmos does not exist. On the contrary, observers in relative motion will not agree on which events occur at the same time. Once again, the reason that this conclusion—a bona fide characteristic of the world we inhabit—is so unfamiliar is that the effects are extremely small when the speeds involved are those commonly encountered in everyday experience. If the negotiating table were 100 feet long and the train were moving at 10 miles per hour, platform observers would “see” that the light reached the president of Forwardland about a millionth of a billionth of a second before it reached the president of Backwardland. Although this represents a genuine difference, it is so tiny that it cannot be detected directly by human senses. If the train were moving considerably faster, say at 600 million miles per hour, from the perspective of someone on the platform the light would take almost 20 times as long to reach the president of Backwardland compared with the time to reach the president of Forwardland. At high speeds, the startling effects of special relativity become increasingly pronounced.

The Effect on Time: Part II

It is difficult to give an abstract definition of time—attempts to do so often wind up invoking the word “time” itself, or else go through linguistic contortions simply to avoid doing so. Rather than proceeding